



Knowledge Graphs and Language Models

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Before we start...

Contact

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GitHub Repository

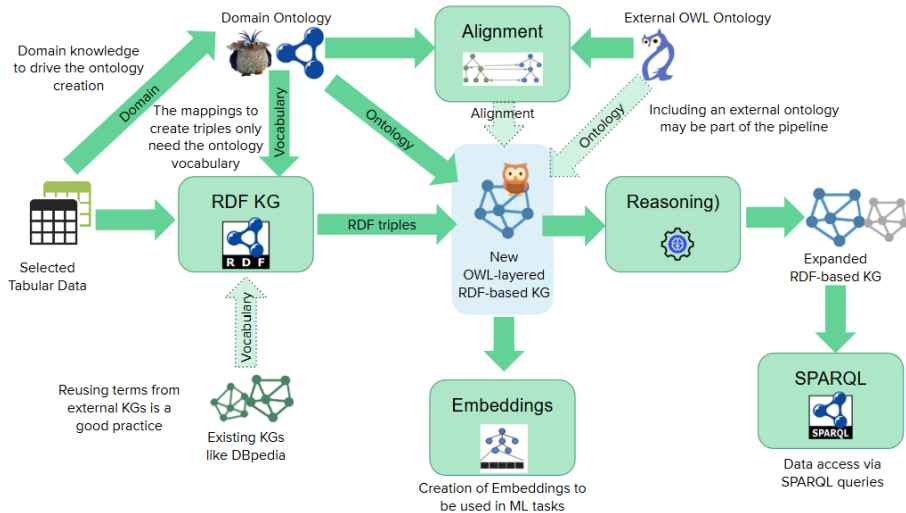
`https://github.com/city-knowledge-graphs/kg-course-valgrai`

Course Organization

1. Introduction to Knowledge Graphs.
2. Modelling Ontologies with OWL.
3. RDFS Semantics, Reasoning and OWL 2 Profiles.
4. From Tabular Data to Knowledge Graphs.
5. Querying Knowledge Graphs and Graph Database Solutions.
6. Knowledge Graphs and Language Models.

The global picture

The global picture



Learning outcomes for today

- Analyse the different options to combine data-driven and knowledge-driven techniques.
- Evaluate the opportunities to integrate large language models with knowledge graphs.
- Apply an ontology embedding system in practice.

Hybrid Learning and Reasoning Systems

Motivation:

- Need of **richer AI** systems, *i.e.*, **semantically sound, explainable, and reliable**.

Gary Marcus. The Next Decade in AI: Four Steps Towards Robust Artificial Intelligence. CoRR abs/2002.06177 (2020)

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- Limitations of KR systems: **maintenance** and **flexibility** in the inference. *e.g.*, Does $C(a)$ hold if?
 - A and $(R \text{ some } B)$ subClassOf C . $A(a)$, $B'(b)$, and $R(a, b)$.

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- **Solution?** Hybrid Learning and Reasoning Systems.

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Hybrid Learning and Reasoning Systems

- Unification of:
 - **statistical** (data-driven) and
 - **symbolic** (knowledge-driven) methods

† Michael van Bekkum et al. Modular Design Patterns for Hybrid Learning and Reasoning Systems: a taxonomy, patterns and use cases. International Journal of Applied Intelligence (2021). <https://arxiv.org/abs/2102.11965>

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Hybrid Learning and Reasoning Systems

- Unification of:
 - **statistical** (data-driven) and
 - **symbolic** (knowledge-driven) methods
- Overview of **patterns** for hybrid systems. †
 - Focus on (1) **Ontology** (knowledge graph) **embeddings** (*e.g.*, OWL2Vec*) and (2) **PLMs/LLMs** as components for a hybrid system

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Statistical Vs Semantic Models

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 - *e.g.*, Ontologies and Knowledge Graphs, Rule-based models.

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 - *e.g.*, Ontologies and Knowledge Graphs, Rule-based models.
- **Hybrid models** combine both.

Types of (logical) inference

– Deductive inference

$$\begin{array}{c} \text{All swans are birds} \\ \text{Daffy is a swan} \\ \hline \text{Therefore, Daffy is a bird} \end{array}$$

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Types of (logical) inference

- **Deductive inference** (this module)

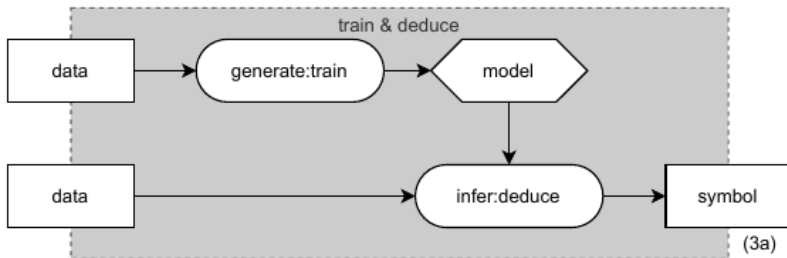
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- **Inductive inference** (a bit today)

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Ross King. **Automatic Science using Robot Scientists.** <https://www.youtube.com/watch?v=Bj1TwqdLBs0>

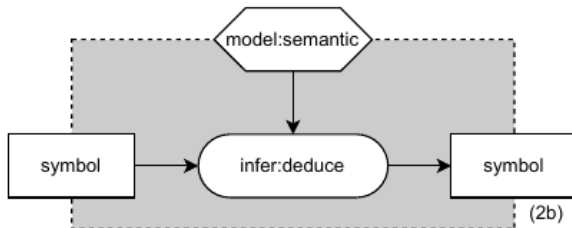
Machine learning pattern (non hybrid)



For example, image classification as in the <http://www.image-net.org/> challenge (symbol = label from WordNet).

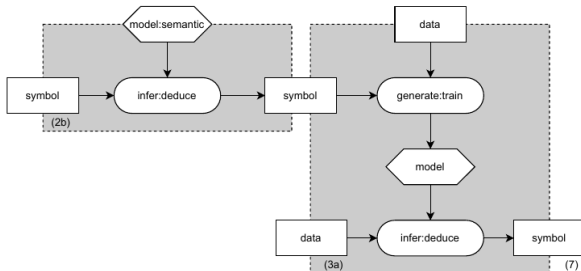
infer:deduce → inductive inference

Semantic model pattern (non hybrid)



- Standard reasoning (*e.g.*, classification, class membership).
 - infer:deduce → deductive inference
- Rule-mining and ontology learning based on symbolic data (*i.e.*, ABox).
 - infer:deduce → inductive inference

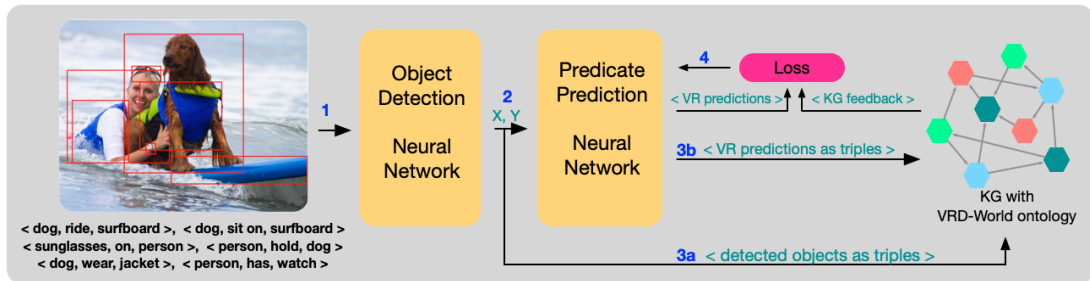
Learning with prior knowledge (i)



- Domain knowledge (e.g., a KG) used to constraint search space during training.
- **Semantic loss function:** impact of the violation of the symbolic knowledge.

A semantic loss function for deep learning with symbolic knowledge. ICML 2018
Logic Tensor Networks. <https://github.com/logictensornetworks/logictensornetworks>

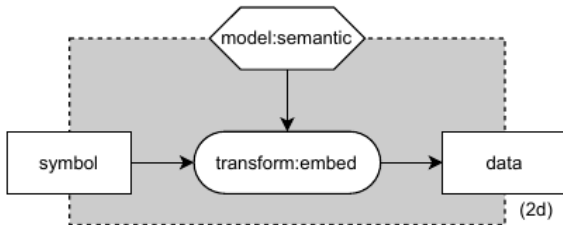
Learning with prior knowledge (ii)



- Penalisation of predictions that violate a constraint in the KG.

D. Herron et al. On the Potential of Logic and Reasoning in Neurosymbolic Systems using OWL-based Knowledge Graphs. Neurosymbolic Artificial Intelligence, 2024.

Knowledge graph embeddings



Symbols are transformed into vectors (*e.g.*, OWL2Vec*)

Knowledge Graph Embedding: A Survey of Approaches and Applications. TKDE 2017
OWL2Vec*: Embedding of OWL Ontologies. Machine Learning journal (2021)

Embeddings (definition)

- An **embedding** is a function that maps a discrete — categorical — variable (*e.g.*, a KG entity/symbol) to a **vector of numbers**.

<https://towardsdatascience.com/neural-network-embeddings-explained-4d028e6f0526>

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- One-hot encoding also assign a vector to categories but...
 - ✗ has as many dimensions as categories $\rightarrow [0 \ 0 \ 0 \ 1 \ 0 \ 0 \dots]$
 - ✗ vectors are not related to each other.

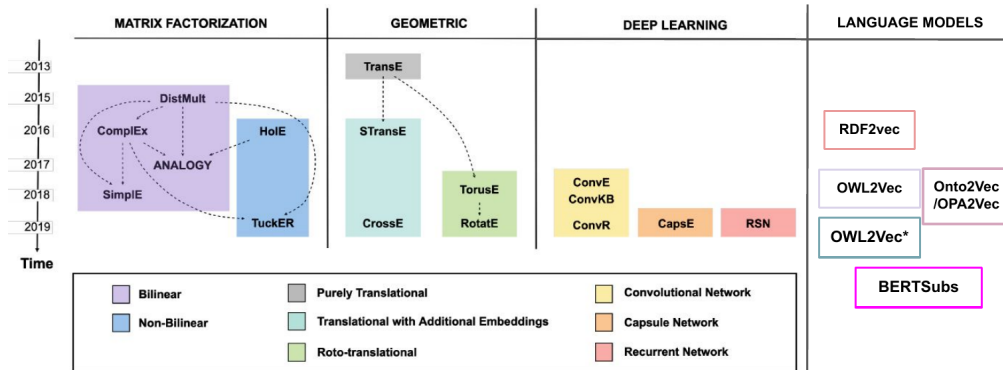
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 - ✗ vectors are not related to each other.
- Embeddings aim at creating **meaningful low-dimensional continuous vectors**.
 - dbr:london $\rightarrow [0.5 \ 0.3 \ 0.8 \ 1.0 \ 0.0 \dots]$

<https://towardsdatascience.com/neural-network-embeddings-explained-4d028e6f0526>

Knowledge graph embeddings (overview of approaches)



Incomplete list of approaches, adapted from: Knowledge Graph Embedding for Link Prediction: A Comparative Analysis. TKDD 2021

Knowledge graph embeddings techniques (self-supervised)

KGE approaches (excluding those based on language models) typically:

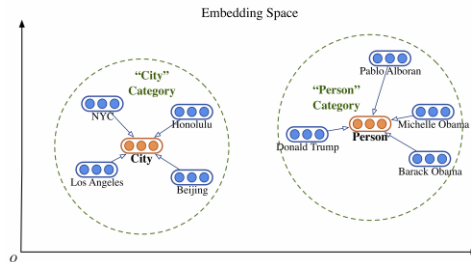
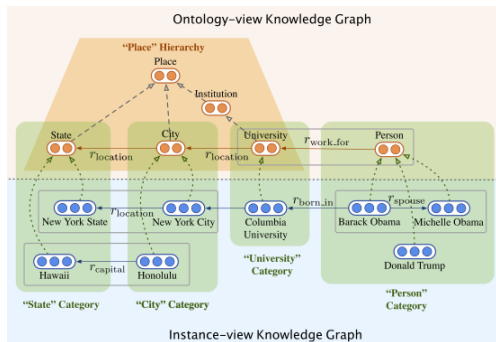
- Receive as input a set of **positive** (the ones in the KG) and **negative triples**.
- Include a **scoring function** that accepts as input the embedding of the elements of a triple (there is an initialization step).

Knowledge graph embeddings techniques (self-supervised)

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- Receive as input a set of **positive** (the ones in the KG) and **negative triples**.
- Include a **scoring function** that accepts as input the embedding of the elements of a triple (there is an initialization step).
- Learn embeddings so that the score for positive triples is maximized while the score for negative triples is minimized (*i.e.*, **loss function**).
- Compute **similar vectors** for similar nodes (*i.e.*, concepts/instances) and edges (*i.e.*, properties).

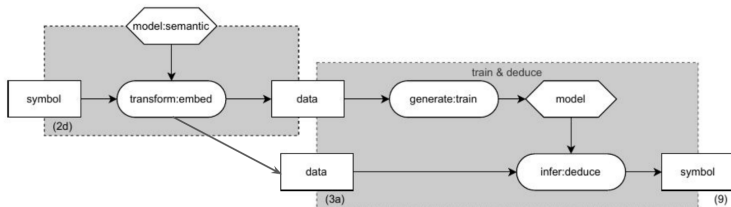
Knowledge graph embeddings (example)



KG Embedding Systems exploit the neighbourhood of an entity to calculate its vector.

Example from: Universal Representation Learning of Knowledge Bases by Jointly Embedding Instances and Ontological Concepts. KDD 2019.

Learning with (knowledge) embeddings (pattern)

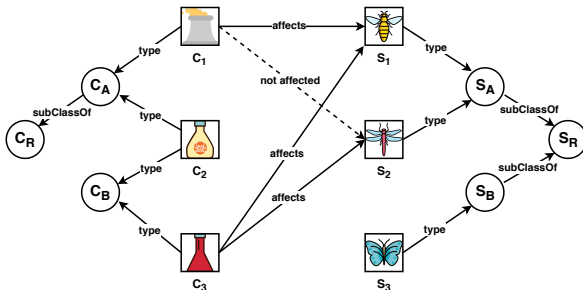


- Applying knowledge graph embeddings in a subsequent classification step.
- Key for zero-shot learning approaches

Prediction of Adverse Biological Effects of Chemicals Using Knowledge Graph Embeddings. Sem Web. 2022.
Knowledge-aware Zero-Shot Learning: Survey and Perspective. arXiv:2103.00070. 2021

Learning with (knowledge) embeddings (example)

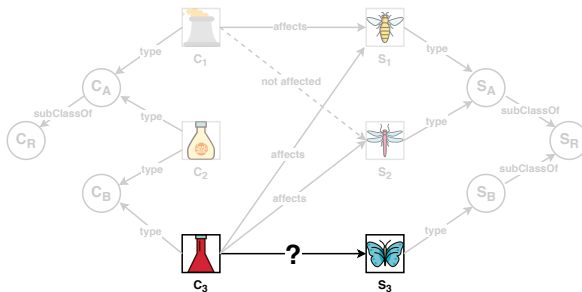
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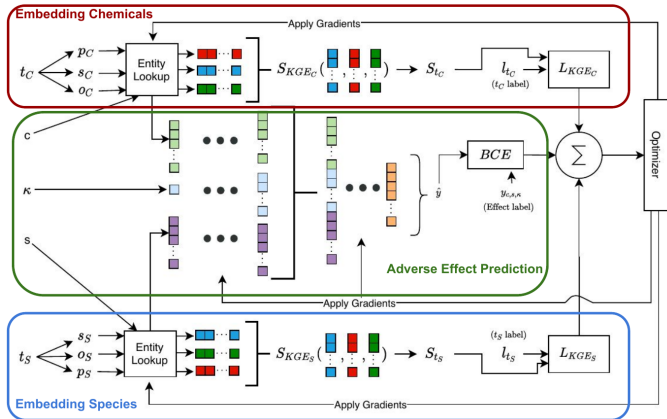
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KGE are critical for unseen chemicals and species. Also useful for explainability.

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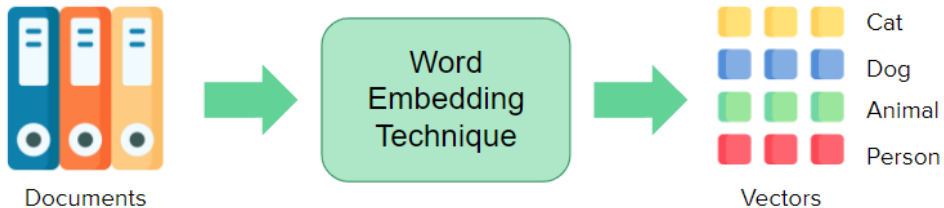


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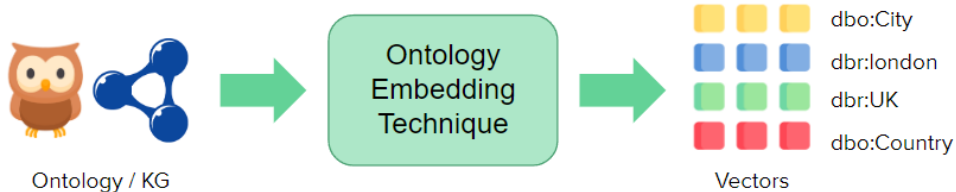
KGs and Language Models: KG Embeddings and beyond

Preliminaries: embeddings

Embedding techniques



Embedding techniques



Word Embedding Techniques (non-contextual)

- **One-hot** embedding.
- Frequency-based Embeddings:
 - **Co-occurrence Matrix.**
 - **TF-IDF** (Term Frequency-Inverse Document Frequency)
 - **GloVe** (Global Vectors for Word Representation)
- **Prediction-based** Embeddings :
 - **Word2Vec** (uses Neural Networks)
 - **FastText** (extends Word2Vec with Subword Information)

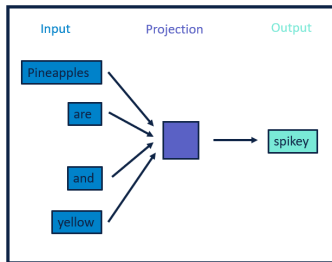
Word2Vec (i)

- **Word2Vec** is a two-layer neural network
- Each unique word is assigned a (**low-dimensional and dense**) vector.
- Two architectural designs: the Continuous Bag of Words (**CBOW**) Model and the Continuous **Skip-Gram** Model.
- Vectors are learned (via an objective function) to capture the **semantic** meaning of the words and **proximity** to other words

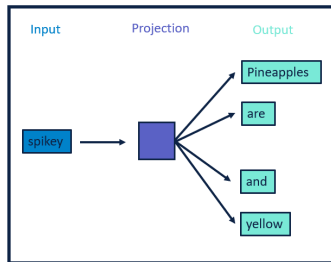
Tomas Mikolov, et al. Efficient Estimation of Word Representations in Vector Space. 2013

Word2Vec (ii)

- CBOW: learns to predict a word given its neighboring words.
- Skip-gram: learns to predict neighboring words given a target word.



CBOW



Skip-gram

[https://swimm.io/learn/large-language-models/what-is-word2vec-and-how-does-it-work\](https://swimm.io/learn/large-language-models/what-is-word2vec-and-how-does-it-work/)

Word2Vec (iii)

Limitations:

- Non contextual embeddings (*bank* vs *river bank*)
- Bias on the embeddings.
- Problem with "out of the vocabulary" words.
- Subwords do not necessarily have similar embeddings (*'end'* and *'endless'*).

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Preliminaries: Contextual embeddings

Transformer-based models (i)

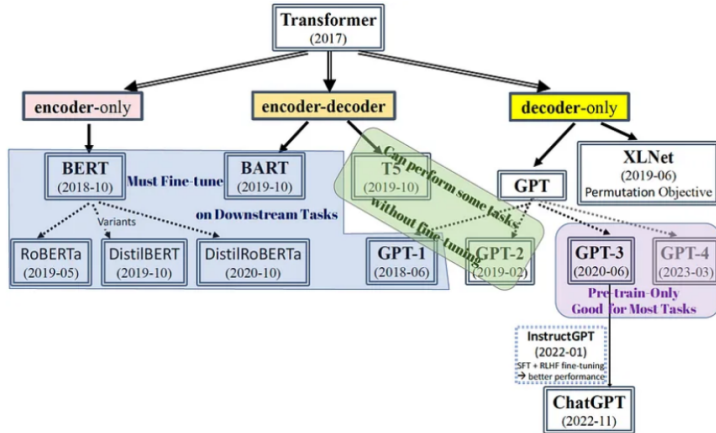
- **Learn contextual embeddings** for each of the words (*i.e.*, a word will have different vectors depending on the context).
- **Pre-trained on (very) large corpora** of text data using unsupervised learning objectives.
- Some require to be **fine-tuned** on specific downstream tasks with labelled data to achieve high performance.
- Have achieved **state-of-the-art performance** on various NLP tasks.

Attention Is All You Need: <https://arxiv.org/pdf/1706.03762.pdf>

Transformer-based models (ii)

- **Three types of models:**
 - Transformer encoders (aka Autoencoders models).
 - Transformer decoders (aka Autoregressive models).
 - Transformer Encoder-Decoder (aka seq2seq models)
- Some examples:
 - **BERT** (Bidirectional Encoder Representations from Transformers),
 - **GPT** (Generative Pre-trained Transformer),
 - **T5** (Text-To-Text Transfer Transformer), and
 - **XLNet**.

Transformers-based models: Taxonomy



<https://medium.com/the-modern-scientist/an-in-depth-look-at-the-transformer-based-models-22e5f5d17b6b>

Transformer-decoder or AR: GPT-based models

- Used by well-known **GPT models**.
- Use the context to predict the **likelihood of the next word**.
- A **deep neural network** (billions of parameters) is trained to model these conditional distributions.

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- A **deep neural network** (billions of parameters) is trained to model these conditional distributions.
- Only trained to encode a **uni-directional context** (either forward or backward).
- Shown impressive potential for **text generation**.
- Harder to fine-tune, but ready to be used in **zero-shot/few-shot** via prompting (scenarios).

AR - Transformer-decoder

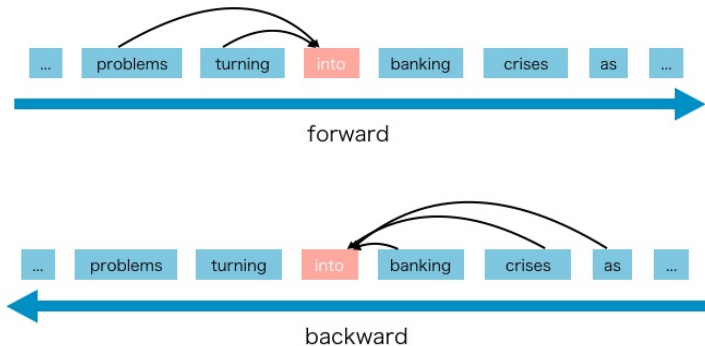


Image from: <https://aman.ai/primers/ai/autoregressive-vs-autoencoder-models/>

Transformer-encoder or AE: BERT-based models (i)

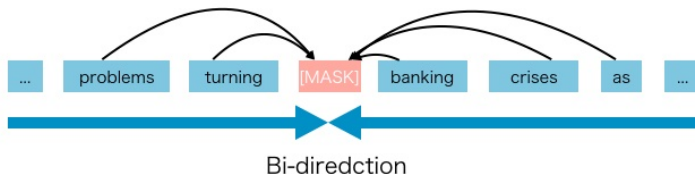
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- **Deep neural network** architecture with million of parameters.
- Pre-training aims to reconstruct the original data from corrupted input (*e.g.*, symbol [MASK]) → **masked language model**.
- Shown impressive performance (after **fine-tuning**) in downstream **text classifications** tasks: spam detection, sentiment analysis, topic categorisation, language detection.

Transformer-encoder or AE: BERT-based models (ii)

- BERT can capture bidirectional context.



- The problem is the introduction of artificial symbols like [MASK].

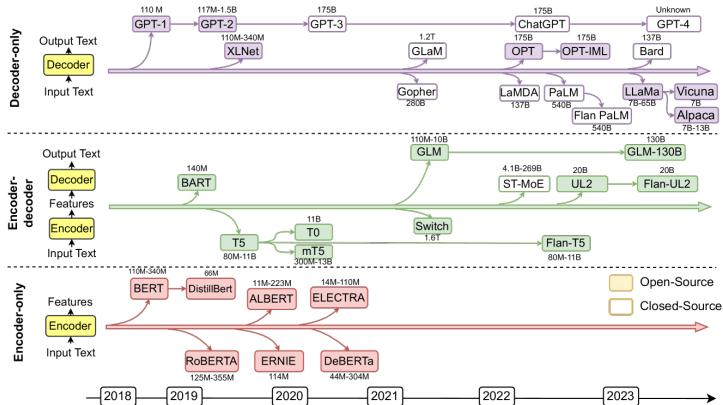
Image from: <https://aman.ai/primers/ai/autoregressive-vs-autoencoder-models/>

Encoder-decoder/seq2seq models

- Use both an encoder and a decoder.
- Each task is considered a sequence to sequence conversion/generation.
- Typically used for tasks that require both content understanding (encoder) and generation (decoder). For example, translation.

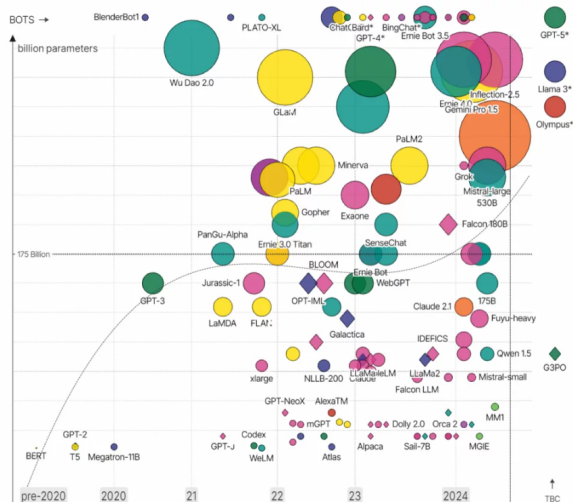
<https://arxiv.org/abs/2308.14149>

LLM Variants (January 2024)



Shirui Pan, et al. Unifying Large Language Models and Knowledge Graphs: A Roadmap. IEEE Transactions on Knowledge and Data Engineering, 2024.

LLM Variants (March 2024)



<https://nestify.io/blog/>

[top-large-language-models/](#)

OpenRouter: Unified Interface

- Interface for the lab session.
- As of March 19, 2026, it provides access to **653 LLM model variants!**
- Easy to use via the OpenAI API.
- `https://openrouter.ai/`

LLMs and KGs: Opportunities and Challenges

Explicit vs. Parametric Knowledge

- **Explicit knowledge:** unstructured knowledge such as text, images and videos; and structured knowledge (*i.e.*, symbolic knowledge) such as knowledge graphs.
- **Parametric knowledge:** refer to the implicit knowledge encoded into the language models' internal parameters (*e.g.*, weights of the neural network).

A key research line is how to transform parametric knowledge into symbolic knowledge. Transformer models can contain **billions of parameters**.

Debate points

- LLMs have shown to generalize from large-scale text corpora.
- LLMs provide plausible answers but not necessarily factually correct.
- LLMs have problems with long-tail knowledge.
- LLMs issues with respect to bias, fairness, copyright violation and misinformation. Hard to “forget” such toxic information from LLMs.
- LLM explainability and interpretability of their predictions.

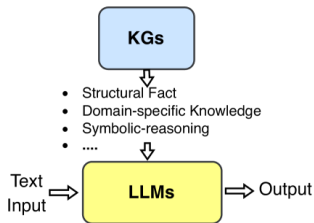
Jeff Pan et al. Large Language Models and Knowledge Graphs: Opportunities and Challenges. TGDK 2023.

Opportunities: LLMs & KGs (i)

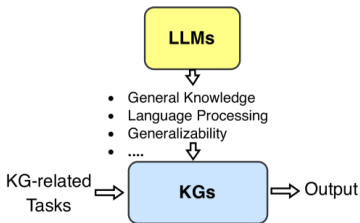
- **Explicit-Knowledge-First:** “LLMs will enable, advance, and simplify crucial steps in the knowledge engineering pipeline so much as to enable KGs at unprecedented scale, quality, and utility.”
- **Parametric-Knowledge-First:** “KGs will improve, ground, and verify LLM generations so as to significantly increase reliability and trust in LLM usage.”

Jeff Pan et al. Large Language Models and Knowledge Graphs: Opportunities and Challenges. TGDk 2023.

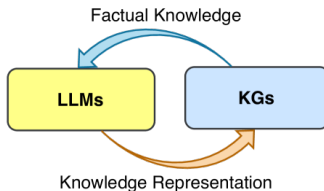
Opportunities: LLMs & KGs (ii)



a. KG-enhanced LLMs



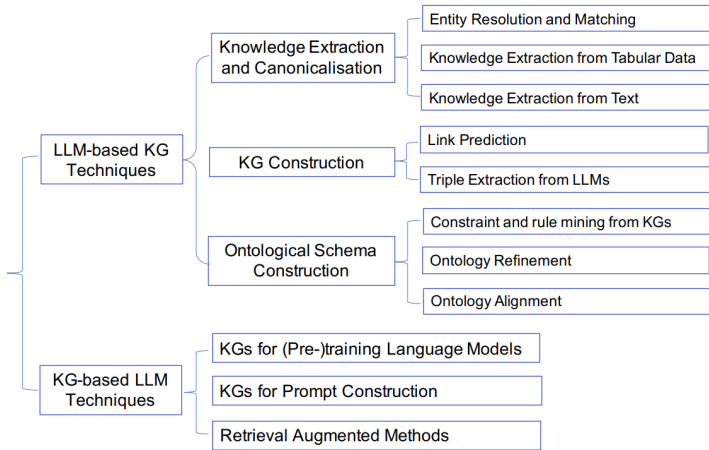
b. LLM-augmented KGs



c. Synergized LLMs + KGs

Shirui Pan, et al. Unifying Large Language Models and Knowledge Graphs: A Roadmap. IEEE Transactions on Knowledge and Data Engineering, 2024.

Opportunities: LLMs & KGs (iii)



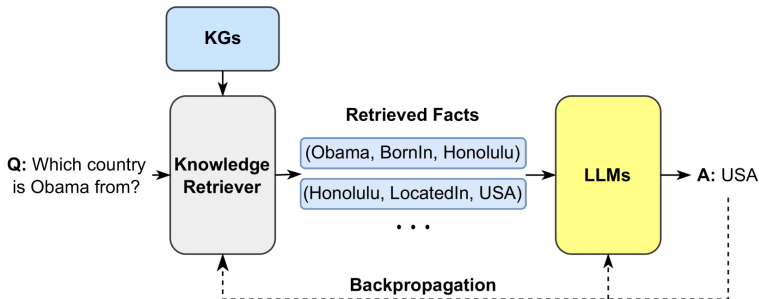
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KGs for LLMs

KG-enhanced LLM Inference

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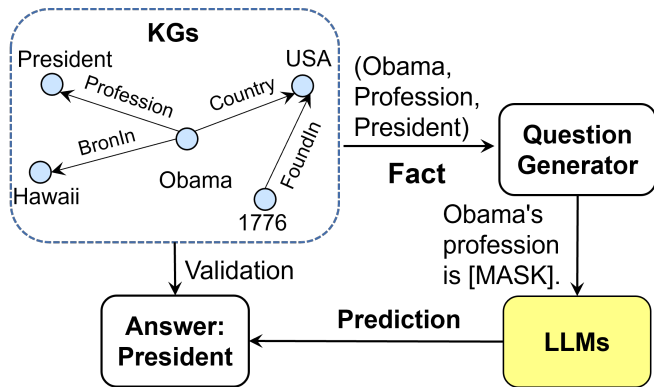
Good to provide the LLMs with fresh/up-to-date facts (without the need of retraining).



Shirui Pan, et al. Unifying Large Language Models and Knowledge Graphs: A Roadmap. IEEE Transactions on Knowledge and Data Engineering, 2024.

KG-enhanced LLM interpretability

KG-enhanced LLM interpretability: Probing

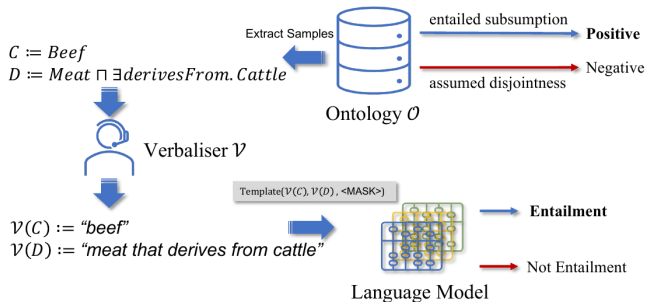


Shirui Pan, et al. Unifying Large Language Models and Knowledge Graphs: A Roadmap. IEEE Transactions on Knowledge and Data Engineering, 2024.

KG-enhanced LLM interpretability: Ontology Inference Probing

OntoLAMA: Language Model Analysis for Ontology Inferencing

- To what extent **PLMs** infer ontology semantics? (e.g., $Beef \sqsubseteq Meat$)



Y. He et al. Language Model Analysis for Ontology Subsumption Inference. ACL findings 2023. <https://arxiv.org/abs/2302.06761>

KG-enhanced LLM interpretability: Ontology Inference Probing

- To what extent **PLMs infer ontology semantics**? (e.g., $C \sqsubseteq D$)
- Natural Language Inference (NLI) for $C \sqsubseteq D$:
 - Premise: “x is a C ” (e.g., “x is a *Beef*”)
 - Hypothesis: “x is a D ” (e.g., “x is a *Meat*”)
- Templates ($Template(C, D, \langle MASK \rangle)$):
 - x is a C , is x a D ? $\langle Mask \rangle$
 - Is it [a/an] C ? $\langle MASK \rangle$, it is [a/an] D (used in paper)

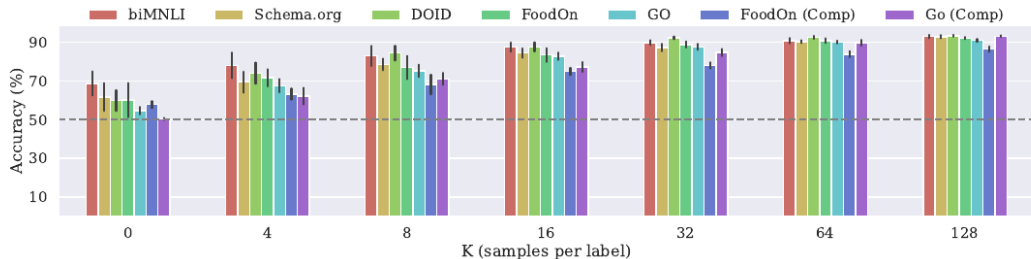
(*) C and D represent labels for atomic concepts or the verbalization for complex concepts.

Y. He et al. Language Model Analysis for Ontology Subsumption Inference. ACL findings 2023. <https://arxiv.org/abs/2302.06761>

KG-enhanced LLM interpretability: Ontology Inference Probing

OntoLAMA: Language Model Analysis for Ontology Inferencing

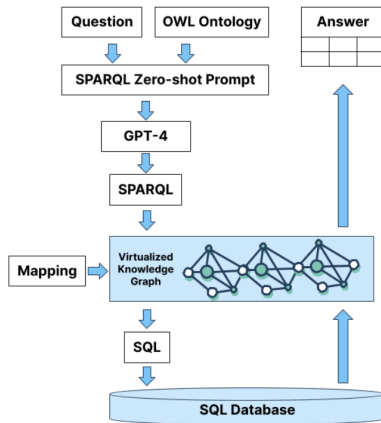
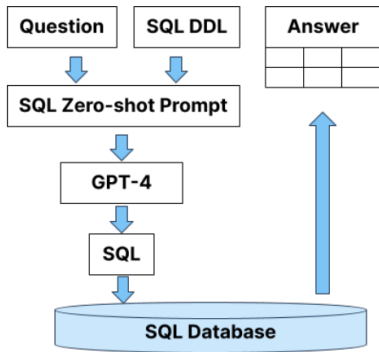
- To what extent **PLMs infer ontology semantics**? (e.g., $Beef \sqsubseteq Meat$)
- **Prompt-based Inference** using RoBERTa in a **K-shot** setting.



Y. He et al. Language Model Analysis for Ontology Subsumption Inference. ACL findings 2023. <https://arxiv.org/abs/2302.06761>

KG-enhanced LLM Question Answering

KGs and LLMs for Question Answering (i)



Juan Sequeda, Dean Allemang, Bryon Jacob: A Benchmark to Understand the Role of Knowledge Graphs on Large Language Model's Accuracy for Question Answering on Enterprise SQL Databases. 2023 <https://arxiv.org/abs/2311.07509>

KGs and LLMs for Question Answering (ii)

	w/o KG (SQL)	w/ KG (SPARQL)	Improvement
All Questions	16.7%	54.2%	37.5%
Low Question/Low Schema	25.5%	71.1%	45.6%
High Question/Low Schema	37.4%	66.9%	29.5%
Low Question/High Schema	0%	35.7%	35.7%
High Question/High Schema	0%	38.5%	38.5%

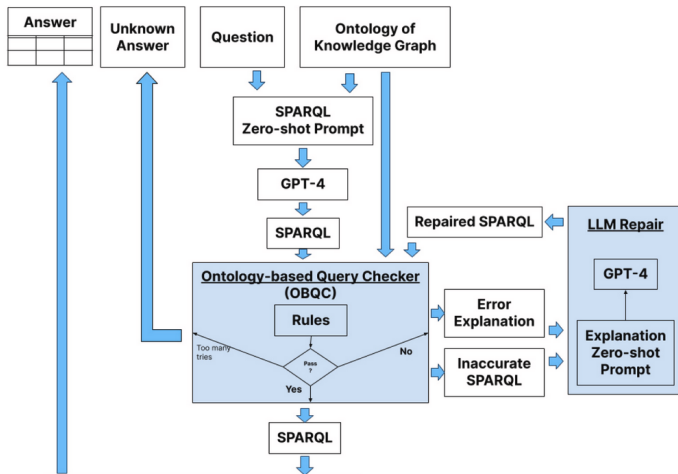
Juan Sequeda, Dean Allemang, Bryon Jacob: A Benchmark to Understand the Role of Knowledge Graphs on Large Language Model's Accuracy for Question Answering on Enterprise SQL Databases. 2023 <https://arxiv.org/abs/2311.07509>

Juan Sequeda, Dean Allemang, Bryon Jacob: Knowledge Graphs as a source of trust for LLM-powered enterprise question answering. J. Web Semant. 85 (2025)

Keynote Turing IG on KGs (I): <https://github.com/turing-knowledge-graphs/meet-ups/blob/main/agenda-7th-meetup.md>

Keynote Turing IG on KGs (II): <https://github.com/turing-knowledge-graphs/meet-ups/blob/main/agenda-9th-meetup.md>

KGs and LLMs for Question Answering (iii)

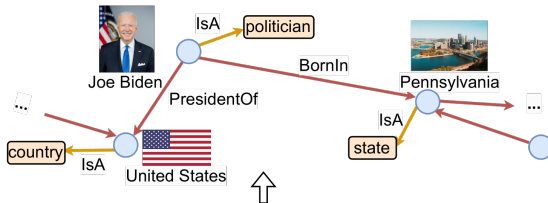


LLMs for KGs

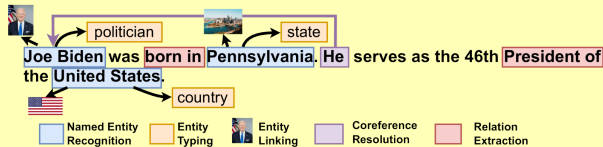
LLM-Enhanced KG Extraction

Knowledge Extraction from Text

Knowledge Graph



LLM-based Knowledge Graph Construction



Text: Joe Biden was born in Pennsylvania. He serves as the 46th President of the United States.

Knowledge Extraction from Tabular Data

Answer the question based on the task below. If the question cannot be answered using the information provided answer with "I don't know".

Task: Classify the columns of a given table with only one of the following classes that are separated with comma: description of event, description of restaurant, postal code, region of address ...

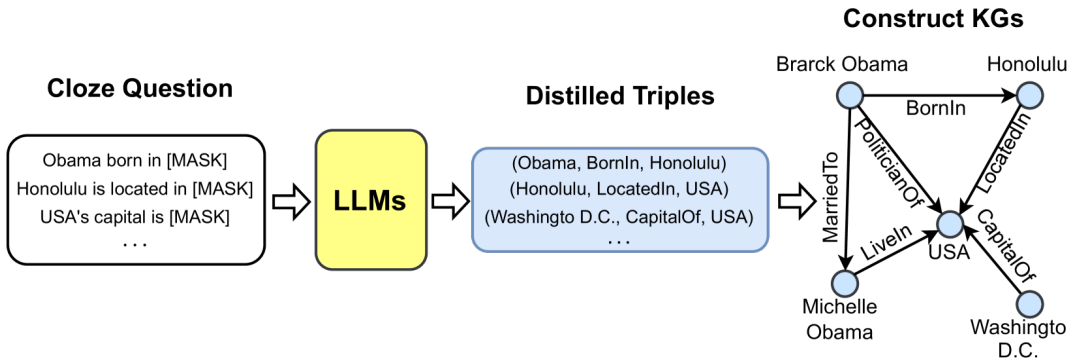
Table: Column 1 || Column 2 || Column 3 || Column 4 \n
Friends Pizza ||2525|| Cash Visa MasterCard || 7:30 AM\n
Class:

name of restaurant, postal code, payment accepted, time

Keti Korini, Christian Bizer. Column Type Annotation using ChatGPT. VLDB Workshops 2023

LLM-Enhanced KG Completion

LLM-Enhanced KG Completion

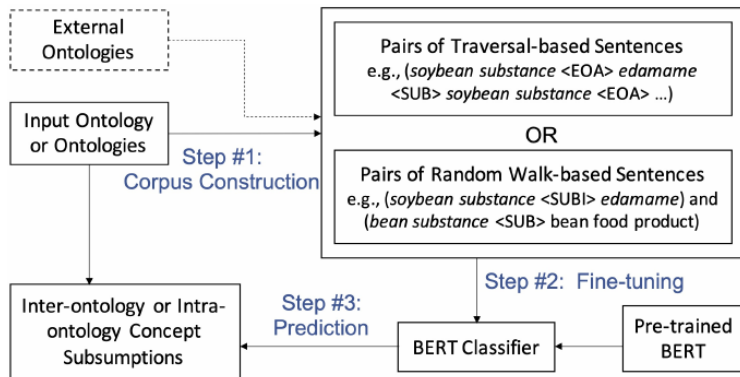


Similar to the probing case but to obtain fresh triples.

Shirui Pan, et al. Unifying Large Language Models and Knowledge Graphs: A Roadmap. IEEE TKDE, 2024.

BERTSubs embeddings for ontology subsumption

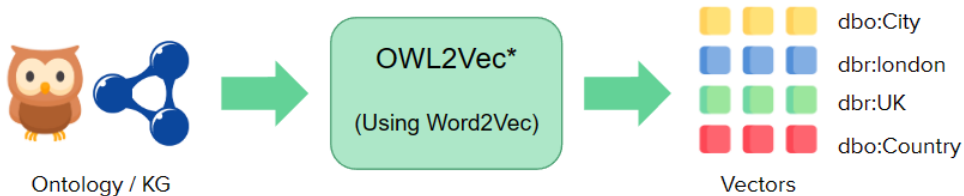
BERTSubs fine-tunes a pre-trained BERT model for ontology subsumption prediction.



J. Chen et al. Contextual Semantic Embeddings for Ontology Subsumption Prediction. World Wide Web Journal 2023

Language Models for KG Embeddings (*e.g.*, OWL2Vec*)

OWL2Vec*: ontology embeddings with Word2Vec (i)



OWL2Vec*: ontology embeddings with Word2Vec (ii)

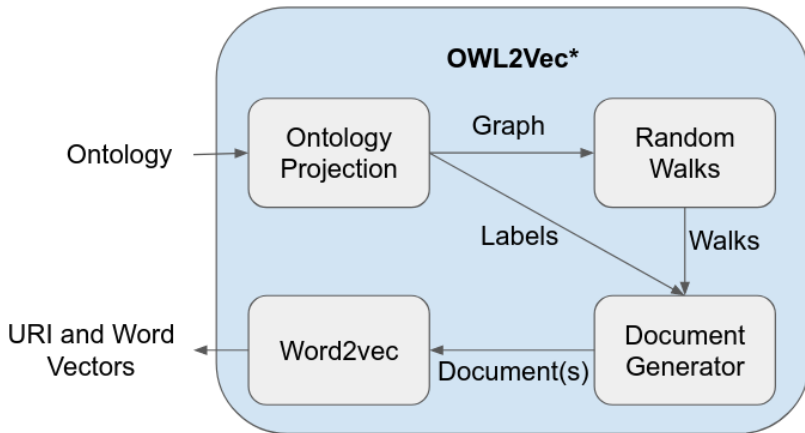
- Inspired by **RDF2Vec**
- Original goal: application to **Ontology Alignment**
- Novelty with respect to RDF2Vec and other KG embedding models:
 - **Ontology axioms.**
 - **Lexical information.**

P. Ristoskiet al.: **RDF2Vec: RDF graph embeddings and their applications**. Semantic Web 2019

O. Magnus Holter, E. Bryhn Myklebust, J. Chen, E. Jiménez-Ruiz: **Embedding OWL Ontologies with OWL2Vec**. ISWC (Satellites) 2019

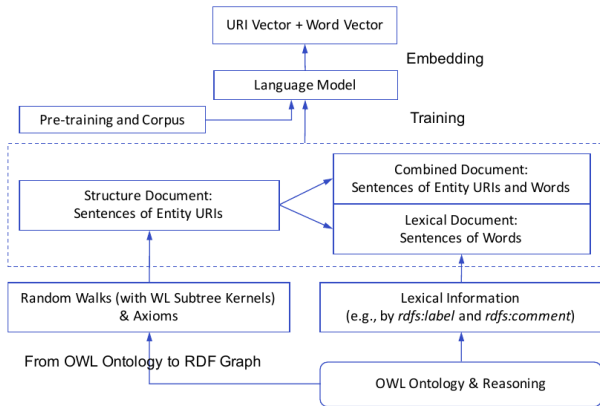
J. Chen, P. Hu, E. Jiménez-Ruiz, O. Magnus Holter, D. Antonyrajah, I. Horrocks: **OWL2Vec*: embedding of OWL ontologies**. Machine Learning 2021

OWL2Vec*: ontology embeddings with Word2Vec (iii)



OWL2Vec*: ontology embeddings with Word2Vec (iv)

- **projects** the ontology into a graph,
- **walks** the graph,
- creates a **corpus of sentences** according to the walking strategies, and
- generates **embeddings** from that corpus using **Word2Vec**.



OWL2Vec*: ontology embeddings with Word2Vec (v)

Projection: Approximation of an OWL 2 ontology into an RDF graph.

Axiom of Condition 1	Axiom or Triple(s) of Condition 2	Projected Triple(s)
$A \sqsubseteq \Box r.D$ or $\Box r.D \sqsubseteq A$	$D \equiv B \mid B_1 \sqcup \dots \sqcup B_n \mid B_1 \sqcap \dots \sqcap B_n$	$\langle A, r, B \rangle$ or $\langle A, r, B_i \rangle$ for $i \in 1, \dots, n$
$\exists r.T \sqsubseteq A$ (domain)	$\top \sqsubseteq \forall r.B$ (range)	
$A \sqsubseteq \exists r.\{b\}$	$B(b)$	
$r \sqsubseteq r'$	$\langle A, r', B \rangle$ has been projected	
$r' \equiv r^-$	$\langle B, r', A \rangle$ has been projected	
$s_1 \circ \dots \circ s_n \sqsubseteq r$	$\langle A, s_1, C_1 \rangle \dots \langle C_n, s_n, B \rangle$ have been projected	
$B \sqsubseteq A$	–	$\langle B, rdfs:subClassOf, A \rangle$ $\langle A, rdfs:subClassOf^-, B \rangle$
$A(a)$	–	$\langle a, rdfs:type, A \rangle$ $\langle A, rdfs:type^-, a \rangle$
$r(a, b)$	–	$\langle a, r, b \rangle$

$\Box$ is one of: $\geq, \leq, =, \exists, \forall$. A, B, B_i and C_i are atomic concepts (classes), s_i, r and r' are roles (object properties), r^- is the inverse of a relation r , a and b are individuals, $\top$ is the top concept.

OWL2Vec*: ontology embeddings with Word2Vec (vi)

Strategies to generate sentences:

- Random walks
- Weisfeiler Lehman (WL) kernel, which assign identifiers to subgraphs and includes them into the walk.

Structure Document Sentences

(vc:Beer, rdf:type, vc:FOOD-4001, vc:hasNutrient, vc:VitaminC_1000)

Lexical Document Sentences

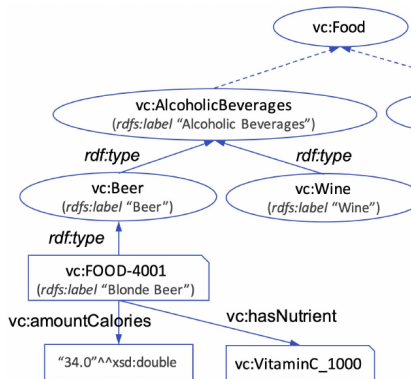
(“beer”, “type”, “blonde”, “beer”, “has”, “nutrient”, “vitamin”, “c”)

Combined Document Sentences

(vc:FOOD-4001, “has”, “nutrient”, “vitamin”, “c”)

OR

(“blonde”, “beer”, “has”, “nutrient”, vc:VitaminC_1000)



OWL2Vec*: ontology embeddings with Word2Vec (vii)

- OWL2Vec* relies on the **Word2vec** as neural **language model**.
- Word2vec learns **embeddings** for all the elements in the documents (*i.e.*, both **words** and **URIs**)

OWL2Vec*: ontology embeddings with Word2Vec (vii)

- OWL2Vec* relies on the **Word2vec** as neural **language model**.
- Word2vec learns **embeddings** for all the elements in the documents (*i.e.*, both **words** and **URIs**)
- The embeddings of the ontology entities can be calculated via their **URI embedding** or via the **word embeddings** of their labels.
 - The URI `vc:FOOD-4001` (Blonde Beer) has a vector.
 - As well as the words ‘blonde’ and ‘beer’.

Rule Mining Using LLMs

A Rule Mining Case with ChatGPT 3.5: uncle



You

Please list three rules that could help to infer one person's uncle. And output it in as datalog rules.

A Rule Mining Case with ChatGPT 3.5: uncle



You

Please list three rules that could help to infer one person's uncle. And output it in as datalog rules.

1. Uncle Rule based on Sibling Relationship:

datalog

 Copy code

```
uncle(X, Y) :- sibling(X, Z), parent(Z, Y), male(X).
```


A Rule Mining Case with ChatGPT 3.5: uncle



You

Please list three rules that could help to infer one person's uncle. And output it in as datalog rules.

1. Uncle Rule based on Brother's Child Relationship:

datalog

 Copy code

```
uncle(X, Y) :- brother(X, Z), parent(Z, Y), male(X).
```

A Rule Mining Case with ChatGPT 3.5: uncle



You

Please list three rules that could help to infer one person's uncle. And output it in as datalog rules.

1. Uncle Rule based on Marriage:

datalog

 Copy code

```
uncle(X, Y) :- married(X, Z), sibling(Z, W), parent(W, Y), male(X).
```

A Rule Mining Case with ChatGPT 3.5: chemical exposure



You

Please list rules that could help to infer chemicals affecting species. And output it in as datalog rules. Use short predicate names.

A Rule Mining Case with ChatGPT 3.5: chemical exposure



You

Please list rules that could help to infer chemicals affecting species. And output it in as datalog rules. Use short predicate names.

1. Chemical Exposure Rule based on Direct Observation:

datalog

 Copy code

```
exposure(S, C) :- observed(S, C).
```

A Rule Mining Case with ChatGPT 3.5: chemical exposure



You

Please list rules that could help to infer chemicals affecting species. And output it in as datalog rules. Use short predicate names.

1. Chemical Exposure Rule based on Habitat:

datalog

 Copy code

```
exposure(S, C) :- inhabits(S, H), hazard(C, H).
```

A Rule Mining Case with ChatGPT 3.5: chemical exposure



You

Please list rules that could help to infer chemicals affecting species. And output it in as datalog rules. Use short predicate names.

1. Chemical Exposure Rule based on Food Chain:

datalog

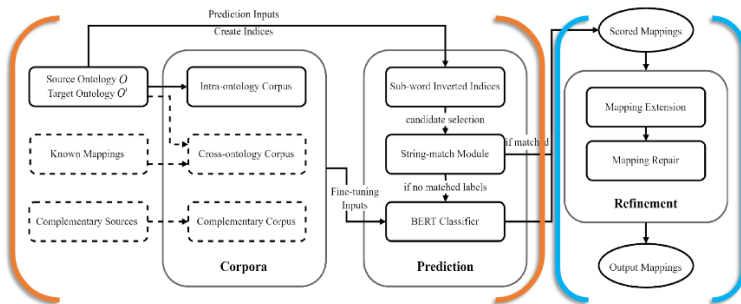
 Copy code

```
exposure(S, C) :- consumes(S, P), exposure(P, C).
```

LLMs for Ontology Alignment

BertMap: Bert-based Ontology Alignment

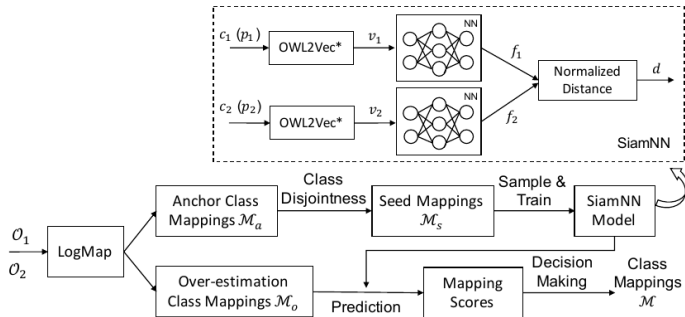
BertMap: fine-tunes BERT with (1) ontology entity synonyms and non-synonyms (unsupervised), and optionally with (2) example mappings (semi-supervised).



Yuan He et al: BERTMap: A BERT-Based Ontology Alignment System. AAAI 2022: 5684-5691.

OWL2Vec*: application to ontology alignment

- LogMap + OWL2Vec* + ML = LogMap-ML
- Self-supervised ontology matching



J. Chen, E. Jiménez-Ruiz et al. Augmenting Ontology Alignment by Semantic Embedding and Distant Supervision. ESWC 2021

OWL2Vec*: application to ontology alignment (ii)

Method	Mappings #	Precision	Recall	F1 Score
LogMap ^{anc}	139	0.892	0.479	0.629
LogMap ^{anc} -ML	157	0.917	0.555	0.691
LogMap	190	0.842	0.618	0.713
LogMap-ML	190	0.881	0.645	0.745
LogMap ^{oei}	198	0.843	0.645	0.731
LogMap ^{oei} -ML	197	0.875	0.665	0.756
AML ^{oei}	220	0.827	0.703	0.760
AML ^{oei} -ML	222	0.842	0.723	0.778

- Results for the OAEI Conference track
- The architecture can be integrated with other OA systems (e.g., AML).

LLMs for Ontology Alignment (i)

- **OntoLAMA/probing** setting applied to inter-ontology subsumptions
- Key: successfully include **context in the prompts**.

James Boyd. MSc Data Science @ City. Investigating OWL Ontology Alignment With Language Models.

LLMs for Ontology Alignment (i)

- **OntoLAMA/probing** setting applied to inter-ontology subsumptions
- Key: successfully include **context in the prompts**.
- Potential templates:
 - *The source entity is C, the target entity is D. Are the concepts equivalent? <MASK>*

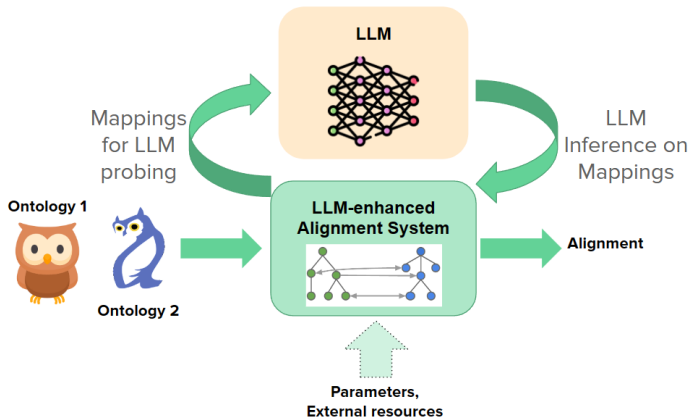
James Boyd. MSc Data Science @ City. Investigating OWL Ontology Alignment With Language Models.

LLMs for Ontology Alignment (i)

- **OntoLAMA/probing** setting applied to inter-ontology subsumptions
- Key: successfully include **context in the prompts**.
- Potential templates:
 - *The source entity is C , the target entity is D . Are the concepts equivalent? <MASK>*
 - *The source entity is $[a/an] C$, a type of C' , the target entity is $[a/an] D$, a type of D' . Are the concepts equivalent? <MASK>*

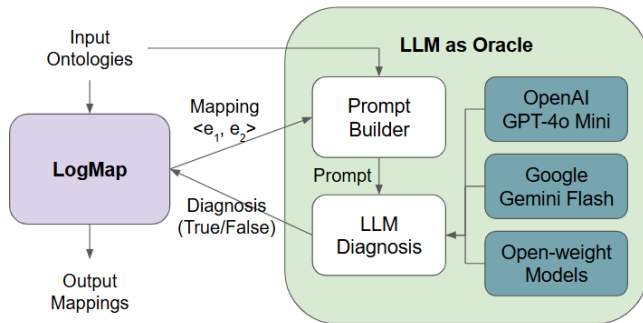
James Boyd. MSc Data Science @ City. Investigating OWL Ontology Alignment With Language Models.

LLMs for Ontology Alignment (ii)



Sviatoslav Lushnei, Dmytro Shumskyi, Severyn Shykula, Ernesto Jiménez-Ruiz, and Artur Garcez. **Large Language Models as Oracles for Ontology Alignment**. In 19th EACL, 2026

LLMs for Ontology Alignment (iii)



Ernesto Jiménez Ruiz, Bernardo Cuenca Grau, Yujiao Zhou and Ian Horrocks. **Large-scale Interactive Ontology Matching: Algorithms and Implementation**. In the 20th European Conference on Artificial Intelligence (ECAI 2012).
Sviatoslav Lushnei, Dmytro Shumskyi, Severyn Shykula, Ernesto Jiménez-Ruiz, Artur d'Avila Garcez: **Large Language Models as Oracles for Ontology Alignment**. arXiv CoRR abs/2508.08500 (2025). Under review for ACL RR.

LLMs for Ontology Alignment (iv)

- We evaluated 6 different types of **ontology-driven prompts**.
- **Best results:** natural language-friendly prompts with synonyms and only one level of context (*i.e.*, parents).

```
We have two entities from different ontologies.
```

```
The first one is "alveolus epithelium", which falls under the category "lung epithelium".
```

```
The second one is "Alveolar_Epithelium", also known as "Lung Alveolar Epithelia", "Alveolar Epithelium", "Epithelia of lung alveoli", which falls under the category "Epithelium".
```

```
Do they mean the same thing? Respond with "True" or "False".
```

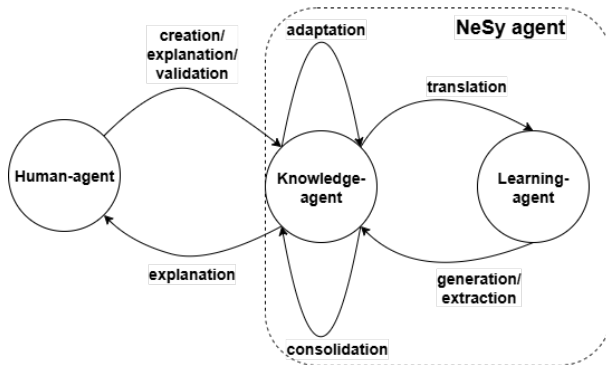

LLMs for Ontology Alignment (v)

Matching task	LogMap	LogMap - Or ^{LLM} _{GF2.0}	LogMap - Or ^{LLM} _{GF2.5}
Mouse-Human	0.880	0.892	0.898
NCIT-DOID	0.869	0.882	0.895
OMIM-ORDO	0.592	0.620	0.626
SNOMED-FMA.body	0.607	0.622	0.632
SNOMED-NCIT.neoplas	0.691	0.694	0.701
SNOMED-NCIT.pharm	0.711	0.719	0.719
FMA-NCI	0.829	0.845	0.828
FMA-SNOMED	0.710	0.719	0.694
SNOMED-NCI	0.743	0.747	0.751
Average	0.737	0.749	0.749

- LLM-based Oracle has a similar performance to an Oracle with an error rate 20%.
- An Oracle with an error rate of 30% would still be better than LogMap.
- LogMap - Or^{LLM}_{GF2.5} (LogMapLLM) ranked #2 in the OAEI Bio-ML 2025 track (LogMap #5).

Towards NeSy Agents

Towards NeSy agents



Ernesto Jimenez-Ruiz et al. Neuro-Symbolic AI Approaches in Medicine: Opportunities and Challenges (White Paper).
Pascal Hitzler et al. Seminar 25291 on (Actual) Neurosymbolic AI: Combining Deep Learning and Knowledge Graphs Dagstuhl Reports, 2025

Acknowledgements

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- **OWL2Vec* developers** and collaborators: Jiaoyan Chen, Pan Hu, Ole Magnus Holter, Denvar Antonyrajah, and Ian Horrocks.
- **The RAI for Ukraine program** (Center for Responsible AI at NYU) and the mentees: Sviatoslav Lushnei, Dmytro Shumskyi, Severyn Shykula
- Dagstuhl Seminar 25291 on (Actual) Neurosymbolic AI: Combining Deep Learning and Knowledge Graphs
- **James Boyd** (MSc Data Science)
- Referenced papers (images, ideas, etc.).
- Icons from <https://www.flaticon.com/free-icons/>

Laboratory Session

Lab Session

- **Using LLMs via OpenRouter.**
 - Test OpenRouter with the provided codes.
 - Test OpenRouter via its chat interface.
- **Creating embeddings with OWL2Vec*.**
 - Execute OWL2Vec* over the `Pizza` and `FoodOn` ontologies.
 - Compute similarity among words and entities.
 - Perform clustering and visualize results.

OWL2Vec* system

